

Q1:) A MOS differential amplifier consists of an NMOS differential pair (M_1-M_2) with a pmos current-mirror active load (M_3-M_4).

The circuit is biased with a tail current source I_{tail} .

$$V_{DD} = 1.8V, \quad I_{tail} = 200\mu A$$

Assumption:

- * All transistors in saturation
- * Perfect matching between M_1-M_2 and M_3-M_4 .
- * Channel-length modulation included

nmos (M_1, M_2):

$$g_{m,n} = 2mS, \quad r_{o,n} = 50k\Omega$$

pmos (M_3, M_4):

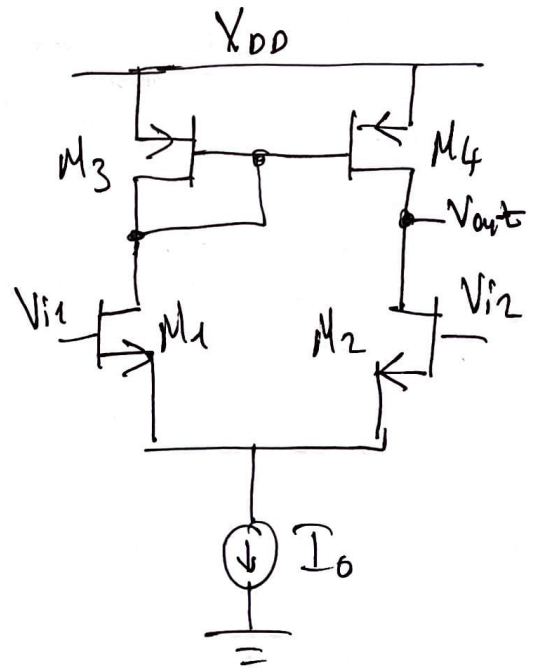
$$g_{m,p} = 1.5mS, \quad r_{o,p} = 40k\Omega$$

The tail current source has a finite output resistance =

$$r_{o,tail} = 200k\Omega$$

Due to device mismatch, the differential pair exhibits an input-referred offset voltage of

$$V_{os} = 2mV$$



a) find the DC current through each nmos. (2)

$$I_D = \frac{I_{tail}}{2} = \frac{200 \mu A}{2} = \boxed{100 \mu A}$$

b) Calculate the single-ended differential-mode gain A_d :

$$r_{out} = r_{o,n} \parallel r_{o,p} = 50 k \parallel 40 k = 22.2 k \Omega$$

$$A_d = g_{m,n} \cdot r_{out}$$

$$A_d = 2 mS \cdot 22.2 k \Omega = \boxed{44.4 V/V}$$

c) Calculate the common-mode gain A_{cm} considering the finite tail resistance.

$$A_{cm} = \frac{r_{out}}{2 r_{o,tail}} = \frac{22.2 k}{2 \cdot 200 k} = \boxed{0.0555 V/V}$$

d) Calculate the CMRR in dB:

$$CMRR = \frac{A_d}{A_{cm}} = \frac{44.4}{0.0555} = 800$$

$$CMRR_{dB} = 20 \log_{10} 800 = \boxed{58 dB}$$

e) Determine the offset voltage

$$V_{out,offset} = A_d \cdot V_{os} = 44.2 \cdot 2 mV = \boxed{88.8 mV}$$

f) How to improve CMRR?

* increase $r_{o,tail}$

* reduce mismatch

* increase $g_{m,n}$

g) Calculate the common-mode gain due to transconductance mismatch. Mismatch $\epsilon = 1\%$.

(3)

$$A_{cm,x} \approx \epsilon A_d$$

$$A_{cm,x} = 0.01 \cdot 40 = \underline{0.4 \text{ V/V}}$$

$$\epsilon = 1\%$$

$A_{cm,x}$ = Common-mode gain due to mismatch

h) Calculate the total common-mode gain.

$$A_{cm,total} = A_{cm,tail} + A_{cm,x}$$

$$= 0.05 + 0.4 = \underline{0.45 \text{ V/V}}$$

i) Calculate the CMRR

$$CMRR = \frac{A_d}{A_{cm,total}} = \frac{40}{0.45} \approx 89$$

$$CMRR_{dB} = 20 \cdot \log_{10} 89 = \underline{39 \text{ dB}}$$